Project Proposal Programming Fundamentals

BATTLESHIP-2D Animation Game

Members

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Project Description

For our 1st semester project, we want to create a **2D battleship project** in which we want to provide a user starting menu in the beginning where he can choose different types of battleships, enemy battleships, and background images. Moreover, the purpose of our game is that our battleship faces different enemies which will move towards our battleship and they are firing towards you, you have to simply dodge the firing of your enemies and if the fire touches you, then your life is decreased which is initially set to 100 percent and after each hit from the fire of the enemy battleship your life will decrease by 5 percent.

Main Objectives

1. Avoid being hit by the firing of enemy battleships.

2. Try to save your lifelines.

3. Try to get highest score by hitting maximum enemy battleships

4. Try to beat the highest score which is set by us in our games and become the highest scorer in the game.

Main Features

1. Main menu

2. Player name

3. Battleship selection

4. Level counter

5. Enemy ships spawn at randomized X-coordinates across the top of the screen.

6. Speed of the enemy battleships will increase with game level

7. Score of the player attained by attacking an enemy's battleships.

8. Selection of enemy battleships' appearance.

9. A dynamic, color-scale health bar and survival time/score tracker rendered directly on screen.

10. Structured screen management, including Main Menu, Background Selection, Gameplay, and Game Over/Restart screens.

Controls

1. Right arrow key to move the battleship right in a row.

2. Left arrow key to move the battleship left in a row.

3. Press Esc to pause the game and show pause menu.

Technology Usage

We will mainly use only **C++** and **SFML** for our project to add graphics and to provide a GUI interface for the user. Moreover, we are using all the concepts being taught by our teachers during our 1st semester such as loops, conditional structures, and variables etc. We are also supposed to use many basic libraries of SFML to include graphics, audio and GUI interface in our project. SFML will be used to provide

users with a starting menu, battleship selection, background selection, and the game score on the screen.

Example/Sample Input

First, the starting menu appears on the screen. Player selects the **“battleship custom 1”** and chooses **“background number 1”**. The game will begin and he hits many enemy objects and tries to dodge the fires of the enemies. But after getting a score of 250, his lifeline becomes zero and the message of “**Game Over**” is shown on the screen. And the menu is shown on the screen from where he can exit or restart the game.

Contact

If anyone wants to contact us regarding any queries or feedback related to our project, they can contact us through our official university emails:

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